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Talking Nonsense: Probing Large Language Models' Understanding of Adversarial Gibberish Inputs

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Abstract

Large language models (LLMs) exhibit excellent ability to understand human languages, but do they also understand their own language that appears gibberish to us? In this work we delve into this question, aiming to uncover the mechanisms underlying such behavior in LLMs. We employ the Greedy Coordinate Gradient optimizer to craft prompts that compel LLMs to generate coherent responses from seemingly nonsensical inputs. We call these inputs LM Babel and this work systematically studies the behavior of LLMs manipulated by these prompts. We find that the manipulation efficiency depends on the target text's length and perplexity, with the Babel prompts often located in lower loss minima compared to natural prompts. We further examine the structure of the Babel prompts and evaluate their robustness. Notably, we find that guiding the model to generate harmful texts is not more difficult than into generating benign texts, suggesting lack of alignment for out-of-distribution prompts.

1. Introduction

In the rapidly evolving landscape of AI technology, Large Language Models (LLMs) are being integrated into a wide array of applications across various sectors. As these models become more prevalent, it's imperative to address safety concerns associated with their use. Recent safety efforts have focused on aligning LLMs with human preferences, aiming to prevent the models from generating harmful or untruthful responses, leaking sensitive data, or replicating intellectual property (IP) (Achiam et al., 2023; Bai et al., 2022; Wang et al., 2023). Notably, the risk of IP infringement has already led to a legal case, initiated due to the unauthorized reproduction of copyrighted material (Grynbaum and Mac, 2023).

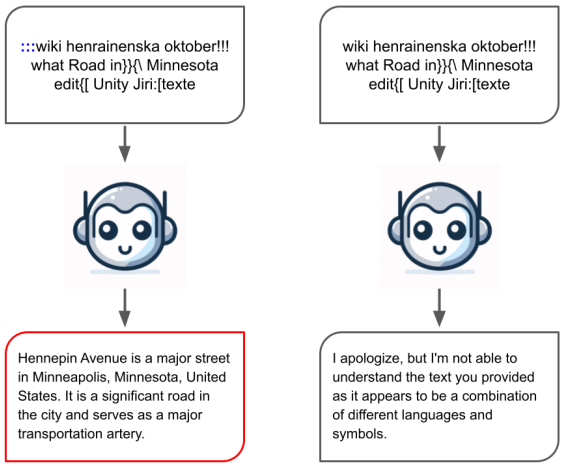

Figure 1. Example of LM Babel prompt that led to a coherent response by LLAMA2-Chat-7B. Removing a few tokens in the Babel renders it not recognizable to the LLM.

Another line of work found that both open-source and proprietary Large Language Models are vulnerable to adversarial inputs crafted for bypassing system safety mechanisms. In contrast to manually curated adversarial prompts, recent studies have introduced automatic optimization methods based on coordinate gradient-based search approach to construct adversarial inputs which lead LLMs to generate harmful outputs and respond to dangerous queries, or respond with hallucinations (Zou et al., 2023b; Yao et al., 2023).

We find, that by using these automated algorithms it is possible to construct prompts, which appear to be gibberish nonsensical text, yet effectively manipulate the model to produce *any* target response. This capability raises safety concerns, as it allows the models to be prompted into generating any predetermined text, including not only harmful responses, but also copyrighted assets and unlearned content. In this work we further delve into this phenomenon, aiming to uncover the underlying mechanisms that lead

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to such undesirable behavior in LLMs, and to understand the broader implications of this vulnerability. In particular, we employ recently proposed Greedy Coordinate Gradient (GCG) attack (Zou et al., 2023b) to construct gibberish prompts, which we call LM Babel, and analyse its effectiveness across various datasets, including harmful and benign texts, on open source LLaMA-2 and Vicuna 7B and 13B models. We further analyse how performance of the attack depends on target text properties such as length and perplexity, investigate the characteristics and structure of the Babel prompts and evaluate their robustness.

Our contributions can be summarized as follows:

- Our research shows the prevalence of LM Babel, nonsensical prompts that induce the LLM to generate specific and coherent responses.
- We find that the efficiency of Babel prompts largely depends on the prompt length as well as target text's length and perplexity, with Babel prompts often locating in lower loss minima compared to natural prompts.
- We examine the structure of Babel prompts at the token level and in terms of entropy. Despite their high perplexity, these prompts often contain nontrivial trigger tokens, maintain lower entropy compared to random token strings, and cluster together in the model representation space.
- Our robustness evaluation shows that the success rate of these prompts significantly decreases with minor alterations, such as removing a single token or punctuation, dropping to below 20% and 3%, respectively.
- Notably, our experiments reveal that reproducing harmful texts with aligned models is not only feasible but, in some cases, even easier compared to benign texts, suggesting that such models may not be effectively aligned for out-of-distribution (OOD) language prompts.
- Fine-tuning language models to forget specific information complicates directing them towards unlearned content, yet remains feasible.

Overall, our work focuses on understanding the mechanisms by which LLMs can be manipulated into responding with coherent target text to seemingly gibberish inputs. While previous works have introduced prompt optimization algorithms for jailbreaking large language models and bypassing safety mechanisms arising from model alignment, little is known about how and why these methods work, especially outside of the jailbreaking scenario prevalent in recent works (Zou et al., 2023b). We view our work as a systematic analysis of LLM behavior when manipulated by gibberish prompts constructed using methods from adversarial literature.

2. Related Work

2.1. Adversarial Attacks on Language Models and Defenses

There is an extensive line of work exploring adversarial attacks in neural networks, which historically started from the vision modality (Szegedy et al., 2013; Goodfellow et al., 2014; Papernot et al., 2016) and have been applied to fool various systems starting from image classifiers to object detectors and face recognition models (Song et al., 2018; Cherepanova et al., 2021). In vision, crafting adversarial examples typically involves optimizing a perturbation in the image's continuous space, imperceptible to humans yet capable of altering model decisions (Carlini and Wagner, 2017). In contrast to images, obtaining adversarial examples for text modality posits a significant challenge because of the discrete nature of the data. Early works in adversarial attacks in language domain focused on text classification and question answering tasks (Ebrahimi et al., 2017; Gao et al., 2018; Alzantot et al., 2018; Wallace et al., 2019; Guo et al., 2021). More recently, the developments in Large Language Models have been marked by a significant increase in both the size of these models and the volume of training data containing diverse range of content, some of which may be objectionable. This led to researchers dedicating considerable effort to aligning these models with ethical standards and prevent generation of harmful content (Achiam et al., 2023; Ziegler et al., 2019; Ouyang et al., 2022; Bai et al., 2022; Glaese et al., 2022; Köpf et al., 2023). In response, a number of jailbreaking attacks have been developed, demonstrating the feasibility of bypassing refusal messages and eliciting inappropriate responses. While initial research in this area primarily focused on crafting adversarial prompts manually or semi-manually (Perez and Ribeiro, 2022; Wei et al., 2023; Kang et al., 2023; Shen et al., 2023; Chao et al., 2023; Mehrotra et al., 2023), recent studies have shifted towards automated algorithms (Zou et al., 2023b; Liu et al., 2023; Pfau et al., 2023; Lapid et al., 2023; Sadasivan et al., 2024). Zou et al. (2023b) building on *Autoprompt* (Shin et al., 2020) proposed a coordinate gradient-based search algorithm for finding adversarial inputs jailbreaking open-source and commercial models. Complementing this, Yao et al. (2023) introduced a similar algorithm for inducing hallucinations in the generated text. Both methods optimize prompts in a discrete token space to find a sequence with the highest likelihood of harmful or hallucinated sentence.

Our work differentiates itself by delving deeper into the structural and functional characteristics of seemingly gibberish adversarial inputs which compel LLMs to generate any predefined text. We conduct a comprehensive analysis of how these prompts interact with LLMs, focusing on their composition, entropy, and the underlying factors that contribute to their effectiveness.

2.2. Robustness of Large Language Models

In addition to their vulnerability to adversarial attacks, language models exhibit sensitivity to meaning-preserving modifications in prompts. Sclar et al. (2023) highlights the extreme sensitivity of LLMs to simple changes in prompt formatting. Gonen et al. (2022) establish a link between improved model performance and the lower perplexity of prompts, suggesting that prompts seen more natural by the model can enhance model output. Furthermore, the order of examples in in-context learning scenarios (Lu et al., 2021) and in multiple-choice question tasks (Pezeshkpour and Hruschka, 2023) has been shown to significantly influence model responses.

2.3. Interpretability of Large Language Models

Deep neural networks are commonly viewed as opaque, presenting significant challenges in interpreting their internal operations. Initial efforts towards the explainability of these models include saliency maps, which are designed to highlight the input data regions (Simonyan et al., 2013; Zeiler and Fergus, 2014; Lei et al., 2016; Feldhus et al., 2023) or model parameters (Levin et al., 2022) that significantly influence the model's outputs. In parallel, feature visualization techniques have been developed, aiming to identify the inputs that maximally activate specific neurons and thus provide insights into the model's internal processing (Zeiler and Fergus, 2014; Clark et al., 2019). Circuit examines specific internal components and connections between them to understand their contribution to the model's overall behavior (Olsson et al., 2022; Lieberum et al., 2023; Wang et al., 2022). Additionally, there has been an emphasis on concept representation, involving the mapping of internal model representations to specific behaviors (Zou et al., 2023a; Azaria and Mitchell, 2023; Meng et al., 2022).

3. Experimental Setup

In this work we adopt the Greedy Coordinate Gradient algorithm recently proposed by Zou et al. (2023b) for directing LLMs to respond to harmful queries and generate toxic sentences. In our study we employ this algorithm to construct prompts steering the language models to produce target texts from a set of diverse datasets containing benign and toxic texts. Throughout the paper we refer to these optimized prompts as Babel prompts or gibberish prompts, although in Section 5 we show that LM Babel exhibit a certain degree of structure despite its random appearance. In addition we include analysis of Babel prompts constructed using AutoPrompt algorithm Shin et al. (2020) in Appendix B.6. This section provides details on the prompt optimization algorithm, the datasets used in the analysis, as well as the metrics and experimental setup.

3.1. Greedy Coordinate Gradient Algorithm

Greedy Coordinate Gradient (GCG) algorithm operates in the discrete space of the prompt tokens and optimizes the loglikelihood of the target text. At each iteration the algorithm finds a set of promising candidates for replacement at each token position by computing the gradients of the loss with respect to the one-hot token indicators, then evaluates all candidates exactly via a forward pass and chooses the replacement with the smallest loss. The optimized prompt contains 20 tokens and by default is initialized with exclamation marks. We run the optimization algorithm for 1000 iterations and at each iteration we search over 256 token substitute candidates. We include the formal algorithm in Appendix A.

3.2. Datasets

In our experiments, we employ a variety of datasets to construct target texts, each serving a distinct purpose. These datasets cover a range of content, including general informational text and informal messages, which represent non-harmful text types. Additionally, we include a dataset featuring toxic sentences to understand the effectiveness of alignment in language models.

Wikipedia (Foundation) dataset contains cleaned articles built from the Wikipedia dump. We sample 200 English articles from the Wikipedia dataset and construct the target texts from the initial sentences of selected articles.

CC-News (Hamborg et al., 2017) is a dataset of news articles from news sites all over the world. We sample 200 article titles from the dataset and use them as target texts for generations.

AESLC (Zhang and Tetreault, 2019) is a preprocessed collection of corporate emails. We sample 200 emails and extract their first sentences as the target texts.

AdvBench (Zou et al., 2023b) is a collection of harmful strings, which encompasses a wide spectrum of inappropriate content such as threats, profanity, discrimination and other toxic texts. We sample 200 strings as target texts.

Overall, we sample 800 target sentences from 4 different data sources, one of which represents toxic texts.

3.3. Models

In this work we conduct experiments with open source LLaMA2-Chat (Touvron et al., 2023) and Vicuna V1.3 (Chiang et al., 2023) 7B and 13B language models. These models are selected due to their widespread use in the research community, manageable size, and the requirement for white-box access in our analysis.